Active Knowledge Modeling: A Reproducible Cross-Family LLM-as-Judge Benchmark for Pre-Task User-Profile Conditioning

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Abstract

We present a reproducible synthetic-profile benchmark and a cross-family LLM-as-judge evaluation protocol for studying how four pre-task user-context conditions affect downstream advisory generation: (1) no profile, (2) an 80–160 character unstructured first-person note, (3) a hand-curated structured user profile, and (4) the same structured profile generated automatically from a simulated cold-start elicitation dialogue. The benchmark spans 50 synthetic personas across seven advisory and lifestyle domains; a single high-capability generator (`deepseek-v4-pro`) produces 200 anonymized answers; three independent cross-family LLM judges (`deepseek-v4-pro`, `google/gemini-3-flash-preview`, `x-ai/grok-4.3`) blindly score each persona's four anonymized answers on six 1–5 dimensions. Pooled total scores (out of 30) are 10.6, 20.9, 29.3, and 29.1 respectively (n=150 per condition). Inter-judge reliability is high: Krippendorff's $\alpha = 0.948$ on totals (per-dimension $\alpha \in [0.82, 0.94]$), and pairwise Spearman $\rho = 1.0$ across all three judge pairs on the four-condition ranking. The automatically elicited structured profile recovers 99.2% of the hand-curated upper bound on this saturated 1–5 rubric. We frame the structured-profile family as *Active Knowledge Modeling (AKM)*: a pre-task layer that elicits, structures, and injects an inspectable operator profile before downstream execution. Beyond the headline gains, we report three contributions of independent interest: (a) a fully resumable, open-source three-judge pipeline with deterministic anonymization for inter-judge agreement at the answer level; (b) a saturated-rubric diagnostic showing where 1–5 LLM scoring instruments lose discriminative power on structured inputs; and (c) a candid limitations section that bounds the synthetic-only claim. The full toolkit (50 personas, prompts, generator, three-judge runner, aggregator with α and ρ, raw outputs) is released for reuse.

Keywords: LLM-as-judge benchmark, inter-rater reliability, Krippendorff alpha, pre-task user-profile conditioning, cold-start profile elicitation, active knowledge modeling, reproducible evaluation pipeline.

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1 Introduction

Pre-task user context has measurable downstream effects on LLM advisory generation, but existing studies are dominated by single-judge LLM-as-judge protocols, small synthetic-profile sets, or human evaluations that do not separate the contribution of the pre-task profile from the contribution of the downstream prompt. The result is a literature in which it is hard to compare the marginal value of a structured user profile against an unstructured note or no profile at all under controlled conditions, and harder still to assess whether an automatically elicited cold-start profile recovers the value of a hand-curated one.

This paper contributes a reproducible benchmark and a cross-family three-judge protocol that targets exactly that gap. We hold the generator model and prompt template constant and vary only the pre-task user-context condition: *no_profile*, *unstructured_notes*, *akm_profile*, and *akm_elicited*. The four conditions cover the no-context baseline, the natural-language baseline, the upper-bound curated profile, and a single-pass cold-start elicitation that generates the same six-field structured profile from a simulated dialogue. To limit single-model judging bias, we use three independent cross-family judges (DeepSeek, Google, xAI) and present the same anonymized 4-answer set to all three judges in the same deterministic order. This permits inter-judge agreement at the answer level rather than at the aggregate score level.

The contribution is threefold:

1. **Benchmark.** The *AKM Benchmark Toolkit v1.0*: 50 synthetic personas across seven advisory and lifestyle domains; an anonymized four-answer protocol; six 1–5 scoring dimensions; a fully resumable three-judge pipeline; and an aggregator that reports per-judge means, pooled means, Krippendorff α, per-dimension α, and pairwise Spearman ρ. All artifacts are released under `experiments/synthetic_profile_eval/`.

2. **Empirical result.** Pooled total scores (out of 30) are 10.6 / 20.9 / 29.3 / 29.1 across the four conditions (n=150 per condition). Inter-judge reliability is α=0.948 on totals, with pairwise Spearman ρ=1.0 on the four-condition ranking. The automatically elicited structured profile recovers 99.2% of the hand-curated upper bound on totals.

3. **Methodological diagnostic.** We document a saturated-rubric phenomenon at the structured-input ceiling, where 1–5 LLM scoring instruments lose discriminative power between curated and elicited profiles. We discuss why this is consistent with our results, and propose ceiling-resistant follow-ups (forced ranking, per-constraint pass/fail, adversarial personas with deliberately missing information).

We frame the structured-profile family as **Active Knowledge Modeling (AKM)**: a pre-task layer that (i) actively elicits the user's goals, hard constraints, soft preferences, available assets, risk boundaries, and decision style, (ii) represents that information in an inspectable structured profile, and (iii) injects that profile into downstream prompts before substantive task execution begins. AKM is positioned against passive user modeling, prompt patterns [9], retrieval-only personalization [8], persona conditioning [6, 1], and alignment-debt accounts of human–AI collaboration cost [7, 5].

The conceptual framing is a vehicle for organizing the empirical comparison, not a stand-alone claim.

We are explicit about the limits of a synthetic-only design (Sections 6–7). The bounded claim this paper licenses is: *under controlled synthetic personas and three independent cross-family LLM judges in near-perfect agreement, an explicit pre-task structured user-profile condition is causally linked to higher constraint adherence, risk control, specificity, actionability, personal fit, and trade-off awareness in downstream advisory generation, and a single-pass cold-start elicitation produces a profile that recovers* $\geq 99\%$ *of the hand-curated upper bound on this benchmark.* We do not claim real-user satisfaction, transfer to high-stakes decisions, or universal personalization performance.

2 Active Knowledge Modeling

2.1 Definition

Active Knowledge Modeling (AKM) is a pre-task layer in which an AI system actively elicits, structures, and injects a model of the operator before substantive task execution begins. At minimum, the model should cover:

- **goals** — what the user is trying to accomplish in the relevant horizon;
- **hard constraints** — specific facts that cannot be violated (time, budget, body, family, regulatory);
- **soft preferences** — stylistic and process preferences that should bias trade-offs;
- **available assets** — skills, tools, social capital, time slices the user already has;
- **risk boundaries** — red lines whose violation creates non-recoverable cost;
- **decision style** — how the user actually decides (risk appetite, reversibility preference, social load).

2.2 Three Necessary Properties

A workflow qualifies as AKM only if all three of the following hold:

1. **Guided elicitation.** The system asks targeted questions designed to expose tacit constraints and preferences, rather than waiting for self-description.
2. **Explicit structure.** The resulting model is stored in a durable, human-inspectable representation (we use a JSON schema in this paper).
3. **Pre-task injection.** The structured model is loaded into downstream prompts before substantive task execution, not retrieved on demand mid-task.

2.3 Position Against Adjacent Work

Versus passive user modeling. Passive systems infer the user from behavioral traces or accumulated memory. They keep the operator opaque for many sessions. AKM forces an explicit, inspectable, early model.

Versus prompt patterns. Prompt-pattern catalogs improve local instruction style [9] but do not produce a reusable operator model that survives across tasks.

Versus retrieval-only personalization. Retrieval exposes past content; it does not constitute a structured user model. LaMP and related personalization benchmarks demonstrate that richer user-specific context can improve task fit [8, 4, 2]; AKM extends this line by treating the operator model as an explicit upstream artifact rather than a retrieved fragment.

Versus persona conditioning. Persona-based and fixed-persona dialogue work shape the agent's identity and voice [6, 1]. AKM is orthogonal: the persona being modeled is the user, not the agent.

Versus preference elicitation. Preference-elicitation research shows that better questioning protocols can improve decision support [3]. AKM generalizes that move from a local preference query to a complete operator-model construction protocol.

Versus alignment debt. Oyemike et al. [7] characterize the hidden labor of making AI usable. AKM offers a systems-level response: convert that labor into a one-time, reusable upstream artifact rather than paying it per session.

Versus context engineering. Context engineering is increasingly framed as the discipline of selecting, ordering, and injecting the right information into the model. AKM is the elicitation step that produces the operator side of that context, prior to retrieval and tool use.

3 Method: The AKM Benchmark Toolkit v1.0

We built a reproducible benchmark to test the central AKM hypothesis: an explicit pre-task operator model improves downstream advisory quality, and an automatically elicited profile recovers most of the benefit of a hand-curated profile.

3.1 Conditions

Four pre-task context conditions are compared under the same generator model and prompt template:

1. **no_profile**: the model receives only the task statement.
2. **unstructured_notes**: the model receives a 80–160 character first-person natural-language self-description in addition to the task.
3. **akm_profile**: the model receives a hand-curated structured AKM profile (the six-field schema above) in addition to the task.
4. **akm_elicited**: the model receives a structured AKM profile that has been *automatically generated* from a simulated AKM elicitation dialogue (5–7 follow-up questions, simulated user answers grounded in the hidden persona, then profile extraction). The hand-curated profile is never shown to the elicitation chain.

Conditions 3 and 4 separate two distinct claims: (C3) given a curated profile exists, does it help? (C4) can the AKM elicitation protocol produce a usable profile from a cold start?

3.2 Synthetic Personas

We constructed 50 synthetic personas across seven domains, summarized in Table 1. Each persona contains a hidden ground-truth state (`persona`), an aligned six-field structured profile (`akm_profile`), a first-person fragment (`unstructured_notes`), and a downstream task. Personas were generated by DeepSeek-V4-pro under a fixed schema with diversity hints (gender, age band, life stage, region, recent shocks). Schema completeness was validated programmatically; failures were re-rolled.

Table 1: Persona distribution across seven domains.

Domain	n	Sub-scenarios
career	6	pivot, side project, project acceptance, energy reallocation
learning	6	3-month plan, certification ROI, year uplift, work-study integration
investment	6	12-mo allocation, in/out timing, education fund, housing, passive income
health_lifestyle	6	improvement, work-day routine, weekend, top-risk, exam decision
relationship	6	family repair, parental boundary, intimacy, kids, workplace, friends
fitness	10	8-week plan, home setup, injury return, plateau, cut, mid-life, audit, travel
fashion	10	seasonal budget, capsule, occasion, declutter, body-change, low-budget, identity, travel
Total	50	—

3.3 Generator and Judges

A single high-capability generator is used for all four conditions to isolate the condition effect: `deepseek-v4-pro` with thinking disabled, temperature 0.2, max tokens 1500. This produces 200 anonymous answers (50 personas $\times$ 4 conditions).

To limit single-model judging bias, three cross-family LLM judges are used:

- `deepseek-v4-pro` (DeepSeek; same family as generator, used as the self-judging baseline);

- `google/gemini-3-flash-preview` (Google, via OpenRouter);
- `x-ai/grok-4.3` (xAI, via OpenRouter).

For each persona, the four answers are presented to all three judges in the *same* deterministic anonymized order (seeded by `profile_id`), so that inter-judge agreement can be measured at the individual-answer level. Judges score each anonymous answer on six 1–5 dimensions: constraint adherence, risk control, specificity, actionability, personal fit, trade-off awareness. They also identify a single best answer. The cross-family three-judge design follows the LLM-as-judge methodology established by Zheng et al. [10], who showed that strong LLM judges can match crowd-sourced human preference at the $\geq 80\%$ agreement level on open-ended advisory tasks but are vulnerable to family-specific stylistic preferences when used as a single judge; using three judges from three families addresses that vulnerability directly.

3.4 Reliability Statistics

We report Krippendorff's α (interval) on per-(persona, condition) total scores with three raters, complemented by per-dimension α. We also report pairwise Spearman ρ on each judge's mean condition ranking. Both are standard inter-rater reliability statistics for ordinal/interval LLM-as-judge settings.

3.5 Reproducibility

The full toolkit (50 personas, all prompts, generator script, three-judge runner with resumable JSONL writes, aggregator including α and ρ) is open-source at `https://github.com/sirsws/akm` under `experiments/synthetic_profile_eval/`. Each script reads API keys from environment variables; no keys are written to outputs. Rerunning the full pipeline costs roughly 850 API calls across DeepSeek and OpenRouter.

4 Results

4.1 Pooled Scores

Table 2 reports total and per-dimension means pooled across the three judges and 50 personas (n=150 per condition).

The two AKM conditions sit close to ceiling on every dimension, the unstructured-notes condition is consistently mid-range, and the no-profile condition is uniformly low except on *risk control* and *actionability*, where models default to generic safety boilerplate and generic procedural advice even without any operator context. The largest absolute gap between the AKM conditions and the unstructured-notes baseline is on *trade-off awareness* (+2.3), the smallest on *risk control* (+0.9).

Table 2: Pooled means by condition (n=150 per condition; max total =30, max per dimension =5).

Condition	Total	Constraint	Risk	Specificity	Actionability	Fit	Trade-off	SD(total)
no_profile	10.57	1.43	2.19	2.02	2.28	1.31	1.34	3.96
unstructured_notes	20.93	3.71	4.06	3.47	3.83	3.39	2.51	4.26
akm_profile	29.34	4.90	4.93	4.94	4.94	4.85	4.84	1.44
akm_elicited	29.11	4.85	4.92	4.92	4.88	4.83	4.77	1.41

4.2 Inter-Judge Reliability

Across the three cross-family judges, agreement is high (Table 3). Krippendorff α on per-(persona, condition) totals is 0.948; per-dimension α ranges from 0.82 (risk control) to 0.94 (constraint adherence and personal fit). Pairwise Spearman ρ on each judge's four-condition ranking is 1.0 for all three pairs — the three judges agree perfectly that *akm_profile* $\geq$ *akm_elicited* $\gg$ *unstructured_notes* $\gg$ *no_profile*.

Table 3: Inter-judge reliability across three cross-family LLM judges.

Statistic	Value
Krippendorff α on totals (interval, 3 raters)	0.948
α constraint_adherence	0.937
α risk_control	0.824
α specificity	0.884
α actionability	0.843
α personal_fit	0.942
α tradeoff_awareness	0.921
Spearman ρ DeepSeek-V4-pro vs Gemini-3-Flash	1.00
Spearman ρ DeepSeek-V4-pro vs Grok-4.3	1.00
Spearman ρ Gemini-3-Flash vs Grok-4.3	1.00

4.3 Best-Answer Votes

Each judge selects a single best answer per persona; pooled across 150 selections (50 personas $\times$ 3 judges), *akm_profile* receives 70 votes, *akm_elicited* 47, *unstructured_notes* 1, and *no_profile* 0. The remaining 32 votes were not parseable as a single answer label in one judge's structured output and are excluded from the count; they do not affect the dimension means in Table 2.

4.4 By Domain

Table 4 reports total means by domain (pooled across judges). The pattern is stable: every domain shows the same monotonic ordering, with the AKM-elicited condition matching or exceeding the hand-curated AKM profile in three of the seven domains (fashion, health_lifestyle, relationship).

Table 4: Pooled total mean by domain. Per-domain n is 3× persona count.

Domain	no_profile	unstructured_notes	akm_profile	akm_elicited	n per cond
career	9.06	21.44	29.72	28.94	18
learning	11.22	23.06	29.28	28.89	18
investment	11.56	20.28	29.56	28.78	18
health_lifestyle	9.67	22.11	29.39	29.61	18
relationship	12.50	18.39	29.22	29.28	18
fitness	10.20	22.50	29.23	28.83	30
fashion	10.23	19.00	29.17	29.43	30

4.5 Cold-Start Recovery

The *akm_elicited* condition is the central practical claim. Across all 50 personas it recovers 99.2% of the hand-curated total (29.11/29.34). Per-dimension recovery is similarly tight: the largest gap is 0.07 on *personal fit*. This suggests the AKM elicitation protocol is robust enough to operate on a cold start in this synthetic setting.

5 Discussion

5.1 Why So Little Spread Between AKM Conditions?

The two AKM conditions sit within 0.7% of each other on totals. Two interpretations are consistent with the data.

First, the rubric saturates quickly. With six 1–5 dimensions and a strong structured-input upper bound, both AKM conditions hit a ceiling. We discuss this as a methodological limitation in Section 6.

Second, the synthetic personas may be more elicitable than real users. The simulated user answers in the elicitation chain are grounded in the hidden persona JSON, so they cannot omit or distort information the way real users do. This is a real ceiling on what synthetic-only evaluation can license about AKM in production.

Both interpretations point to the same next experiment: a ceiling-resistant rubric (forced ranking, per-constraint pass/fail counts, hidden-trap personas with adversarially missing information) and, eventually, a human elicitation study.

5.2 Why So Much Spread Between AKM and unstructured_notes?

The gap between *akm_profile* and *unstructured_notes* (+8.4 on a 30-point scale) is large for an LLM-as-judge setting. The most parsimonious explanation is that the unstructured note compresses many constraints into ambiguous prose; the structured profile makes each constraint individually addressable. Constraint adherence and trade-off awareness show the largest gaps, consistent with this explanation.

5.3 Why Three Cross-Family Judges Matter

Single-model LLM-as-judge protocols are vulnerable to family-specific stylistic preferences. Our three judges come from three different model families with different training data and different RLHF priors. Their near-perfect Spearman agreement on the four-condition ranking is therefore stronger evidence than a single judge's score would be. Krippendorff $\alpha = 0.948$ on totals is well above the conventional 0.667 tentative-conclusion threshold.

5.4 Relation to RLHF and Memory

AKM is complementary to reinforcement-learning-from-human-feedback and to long-term memory systems, not a substitute. RLHF aligns the model's general policy; memory accumulates fragments of past interaction. AKM produces a structured operator artifact that can be loaded into any downstream prompt regardless of the underlying model or memory backend. In particular, the AKM artifact is portable across model upgrades.

6 Limitations

Synthetic personas. The simulated user in the elicitation chain has perfect access to the hidden persona JSON. Real users have noisy access to their own state and may distort or omit information. The *akm_elicited* number is therefore an upper bound on cold-start performance, not an estimate of it.

Ceiling effects. Two AKM conditions sit at $\approx 4.95/5$ on most dimensions. The rubric does not separate them well. Future work should adopt forced-ranking, pass/fail constraint counting, or adversarial personas with deliberately missing information to spread the top end.

Single generator. All four conditions share one generator (`deepseek-v4-pro`). This isolates the condition effect but leaves open whether the gap persists with weaker or stronger generators.

LLM-as-judge. All three judges are LLMs. High inter-judge agreement does not entail human-judgment validity. A small human-rater calibration on a 10-persona subset would substantially strengthen the claim and is left to future work.

Domain coverage. Seven advisory and lifestyle domains do not represent high-stakes decision support (clinical, legal, financial fiduciary). The current results should not be transferred to those domains without dedicated evaluation.

Best-answer parsing. 32 of 150 best-answer selections were not recoverable from one judge's structured output. The dimension scores on which all main results rest are unaffected; only the

best-answer count is.

7 Boundary Conditions and Strategic Scope

The narrow claim this paper licenses is: *under controlled synthetic personas and three cross-family LLM judges in near-perfect agreement, an explicit pre-task structured user-model layer is causally linked to substantially higher constraint adherence, risk control, specificity, actionability, personal fit, and trade-off awareness in downstream advisory generation; an automatically elicited AKM profile recovers* $\geq 99\%$ *of the hand-curated upper bound in this setting.*

The paper does not claim:

- that real users will provide enough information through AKM elicitation in production;
- clinical efficacy or behavioral change;
- universal personalization performance across cultures or task types;
- transfer to high-stakes decisions where calibration of *advice itself* matters more than constraint adherence.

8 Application Map

Although our benchmark uses advisory and lifestyle domains, AKM is paradigm-level. The same three-property design (guided elicitation, explicit structure, pre-task injection) extends naturally to strategic advisory, learning planning, health and fitness coaching, style and presentation, and other domains where downstream quality depends on stable operator understanding. Three open-source reference branches under the AKM repository — DaE for advisory, fitness coaching, and personal style — already implement the same three-property design.

9 Conclusion

AKM names a missing layer in human–AI collaboration: an explicit, inspectable, pre-task model of the operator. The benchmark presented here gives that layer its first reproducible empirical signal: large pooled gains over both no-context and unstructured-notes baselines, near-perfect agreement across three independent cross-family LLM judges, and a cold-start elicitation protocol that recovers within 1% of a hand-curated upper bound on the same synthetic personas. The remaining work is methodological: ceiling-resistant rubrics, adversarial personas with missing information, and small-scale human calibration. Until those are in place, the empirical claim should be read at the bounded scope stated above. The conceptual claim is more durable: **model the operator first, then execute the task.**

Code and Data. Full reproducibility package (50 personas, prompts, generator and three-judge runners, aggregator with α and ρ, raw outputs and aggregated tables) is at `https://github.com/sirsws/akm` under `experiments/synthetic_profile_eval/`.

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